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Sem - 8th

Dept - ECE

Sub - Industrial Automation and control

Subj code - PE-EC 802B

Multiple choice questions

① What is the full form of FIR in terms of Signal filtering?

→ Ans ⓐ Finite Impulse Response

② Which frequency is attenuated in a low-pass filter?

→ Ans ⓐ High frequency

③ Which is a Signal generator module?

→ Ans ⓓ EddyNCDT 3301

④ Which Signal is obtained after differentiating the unit Step Signals?

Ans → Unit Step Signal

⑤ Which type of wave can be generated by LM566?

→ Ans ⓒ Square

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Short Answer type questions

① Explain Successive approximation method for analog to digital converter.

→ It is a type of ADC that converts analog signal to digital using binary search method

It contains: comparator, DAC, SAR register, clock.

Working

ⓐ MSB is set to 1

ⓑ DAC converter it to analog value.

ⓒ Comparator checks input voltage with DAC output.

ⓓ If DAC output is less → bit remains 1; if more → bit becomes 0.

ⓔ process continues bit by bit until LSB. It is fast and commonly used in microcontrollers.

② Define Sensor protection circuit. What are the types of sensor protection circuit?

Ans → A sensor protection circuit protects sensors from overvoltage, overcurrent, noise, and short circuits.

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Types

① overvoltage protection (zener diode, TVS diode)

② overcurrent protection (fuse, current limiting resistor)

③ Reverse polarity protection (Diode)

④ Short circuit protection

③ Explain isolation in signal conditioning. Mention different type of isolation Methods.

Ans → Isolation prevents high voltage or noise from damaging circuits.

Types of isolations

① optical isolation (optocoupler)

② Transformer isolation

③ Capacitive isolation

④ Magnetic isolation

④ A 10 bit ADC has a scale of 10.230V, when the digital output is $(11111111\ 1)_2$ the quantization error of the ADC in Millivolt is ?

Ans → given full scale = 10.230V

Digital output = $(1111111111)_2$ = 1023

for 10-bit ADC → total levels = 2^{10} = 1024

Step size (Resolution) = 10.230 / 1024

= 0.00999 V ≃ 0.01V

quantization error = ± (Step size / 2)

= ± (0.01 / 2)

= ± 0.005V

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